

Conditional statements



1

a No	b Yes	c No	d No
e No	f No	g Yes	h No
i No	j No	k Yes	l No
m No	n Yes	o No	p Yes
q Yes	r No	s No	t No
u No	v No	w Yes	x Yes

2

a No	b No	c Yes	d Yes
e No	f No	g Yes	

3

- a** If two lines are parallel, then they never intersect.
- b** If the animal is a cat, then it cannot speak French.

4

- | | |
|--------------|-------------|
| a Yes | b No |
|--------------|-------------|

5

- a** Some birds are not able to fly.
- b** All turtles have claws.
- c** Some Libras are Geminis.
- d** Some bicycles have four wheels.
- e** Some houses are not connected to the internet.

6

- | | |
|--|--|
| a If D , then C . | b If $\sim C$, then $\sim D$. |
| c If $\sim D$, then $\sim C$. | |

7

- a** If an animal has claws, then it is a bear.
- b** If a boomerang is delicious, then it is round.
- c** If a figure is a quadrilateral, then it is a square.

- 8
- a If an animal does not have feathers, then  not a bird.
 - b If a number is not divisible by ten, then it does not end in zero.
 - c If a bamboozle is not loud, then it is not square.

- 9
- a If an animal is not a bird, then it does not have feathers.
 - b If two angles do not form a right angle, then they are not complementary.
 - c If a heffelump is not lost in the woods, then it is not a woozle.

10 Converse

Additional questions

- 11 If p , then q .
- 12 No, a conditional statement does not need to be true.
- 13 The following statements are conditional statements:
- a If a polygon has three sides, then it is a triangle.
 - c If I go to school in Florida, then I live in Orlando.
 - d If it is raining, then we cannot play baseball.
 - e If a parallelogram is not a rhombus, then it is not a square.
- 14
- a Hypothesis: Two angles are vertical
Conclusion: They are congruent
 - b Hypothesis: The sum of a polygon's interior angle measure sum is 180°
Conclusion: The polygon is a triangle
 - c Hypothesis: Two triangles have the same side lengths
Conclusion: They are congruent
 - d Hypothesis: Two lines are parallel
Conclusion: The consecutive interior angles formed by a transversal are supplementary
- 15
- a If I am an artist, then I can paint a picture.



16

- a Converse statements b Contrapositive statements
- c Inverse statements

17

- a False. I can go to any store or online retail website to go shopping. For example, I can go to Southeastern Grocers if I need to go shopping.
- b True
- c True
- d False. Many different people go to schools for work or to visit. For example, if I go to school I could be a teacher.

18

- a Yes, this statement is true.
- b No. The converse statement, "if $x^2 = 4$, then $x = 2$ " is not true because $x = -2$ is also possible.
- c No. The inverse statement, "if $x \neq 2$, then $x^2 \neq 4$ " is not true because $x = -2$ is possible.
- d Yes, the contrapositive is true.

19 If a polygon is a square, then it has four right angles.

The converse of this statement "If a polygon has four right angles, then it is a square" is not true because a rectangle has four right angles and is not necessarily a square.

20

- a If two angles are vertical, then they are congruent.
- b If two angles are not congruent, then they cannot be vertical.

21

- a If you don't take a shot, then you can never score it.
- b If they are bigger, then they will fall harder.
- c If you are in doubt, then tell the truth.